3. $p(x)=\begin{cases}\frac{e^{-\frac{x}{\theta}}}{\theta}, & x\geq 0\\ 0, & x<0\end{cases}$ $\theta>0$

По выборке $n=3$ найдены оценки: $\tilde{\theta}_1=\bar{x}$, $\tilde{\theta}_2=x_{(2)}$

а) Проверить оценки на несмещенность. Исправить, если необх

в) Какая из исправленных оценок более эффективна?

с) Исследовать эти оценки на эффект. с помощью нер-ва Крамера-Рао

$$M\xi=\int_{-\infty}^{+\infty}xp(x)dx=\int_0^{+\infty}\frac{x}{\theta}e^{-\frac{x}{\theta}}dx=\left\{\begin{matrix}x=u & dx=du\\ e^{-\frac{x}{\theta}}dx=dv & -\theta e^{-\frac{x}{\theta}}=v\end{matrix}\right\}=$$

$$=\frac{1}{\theta}\left(x\cdot(-\theta e^{-\frac{x}{\theta}})\Big|_0^{+\infty}+\int_0^{+\infty}\theta e^{-\frac{x}{\theta}}dx\right)=\frac{1}{\theta}\left(-\theta^2e^{-\frac{x}{\theta}}\Big|_0^{\infty}\right)=$$

$$=\frac{1}{\theta}(\theta^2)=\theta$$

$$M\xi^2=\int_{-\infty}^{+\infty}x^2p(x)dx=\int_0^{+\infty}\frac{x^2}{\theta}e^{-\frac{x}{\theta}}dx=\frac{1}{\theta}\left(x^2\left(e^{-\frac{x}{\theta}}(-\theta)\right)+\right.$$

$$\left.+2\theta\int_0^{+\infty}xe^{-\frac{x}{\theta}}dx\right)=\frac{1}{\theta}2\theta^3=2\theta^2$$

$$D\xi=2\theta^2-\theta^2=\theta^2$$

$$\tilde{\theta}_1=\bar{x}=\frac{1}{n}\sum_{i=1}^{n}x_i$$

$$M\tilde{\theta}_1=M\left[\frac{1}{n}\sum_{i=1}^{n}x_i\right]=M\xi=\theta$$

$\Rightarrow \tilde{\theta}_1$ - несмещ

$$D\tilde{\Theta}_1 = D\left[\frac{1}{n}\sum_{i=1}^{n} x_i\right] = \frac{1}{n^2}\sum_{i=1}^{n} D\xi = \frac{\Theta^2}{n}$$

$$\tilde{\Theta}_2 = x_{(2)}$$

$$\mathcal{L}(x) = n\frac{e^{-\frac{x}{\Theta}}}{\Theta}(n-1)e^{-\frac{(n-2)x}{\Theta}}\left(1-e^{-\frac{x}{\Theta}}\right)$$

$$M[x_{(2)}] = n(n-1)\int_0^{+\infty}\frac{x}{\Theta}\left(e^{-\frac{(n-1)x}{\Theta}} - e^{-\frac{nx}{\Theta}}\right)dx =$$

$$= n(n-1)\int_0^{+\infty}\Theta t\left(e^{-(n-1)t} - e^{-nt}\right)dt = n(n-1)\Theta\left(\frac{1}{n-1}\int_0^{+\infty}e^{-(n-1)t}dt -\right.$$

$$\left.-\frac{1}{n}\int_0^{+\infty}e^{-(n-1)t}dt\right) = \left(\frac{n(n-1)}{(n-1)^2} - \frac{n(n-1)}{n^2}\right)\Theta = \Theta\left(\frac{n}{n-1} - \frac{n-1}{n}\right) =$$

$$= \frac{2n-1}{n(n-1)}\cdot\Theta \quad \text{смещ.}$$

$$\tilde{\Theta}_2' = \frac{n(n-1)}{2n-1}\tilde{\Theta}_2 \text{ - несмещ}$$

$$M\tilde{\Theta}_2^2 = n(n-1)\Theta^2\int_0^{+\infty}t^2\left(e^{-(n-1)t} - e^{-nt}\right)dt =$$

$$= n(n-1)\Theta^2\left(\frac{2}{n-1}\int_0^{+\infty}te^{-(n-1)t}dt - \frac{2}{n}\int_0^{+\infty}te^{-nt}dt\right) =$$

$$= n(n-1)\Theta^2\left(\frac{2}{(n-1)^2}\int_0^{+\infty}e^{-(n-1)t}dt - \frac{2}{n^2}\int_0^{+\infty}e^{-nt}dt\right) =$$

$$= \left(\frac{2n}{(n-1)^2} - \frac{2(n-1)}{n^2}\right)\Theta^2 = 2\Theta^2\cdot\frac{3n^2-3n+1}{n^2(n-1)^2}$$

$$D\tilde{\Theta}_2 = \frac{3n^2-3n+1}{n^2(n-1)^2}2\Theta^2 - \frac{4n^2-4n+1}{n^2(n-1)^2} = \frac{2n^2-2n+1}{n^2(n-1)^2}\Theta^2$$

$$D\tilde{\Theta}_2' = \frac{2n^2-2n+1}{4n^2-4n+1}\Theta^2$$

$$D\tilde{\Theta}_1 = \frac{\Theta^2}{n} \qquad D\tilde{\Theta}_2' = \Theta^2 \frac{\left(2 - \frac{2}{n} + \frac{1}{n^2}\right)}{\left(4 + \frac{4}{n} + \frac{1}{n^2}\right)}$$

$$\cancel{D\tilde{\Theta}_2' = \Theta^2 (2n^2} = 2n$$

$\Rightarrow \exists n_0 : \forall n \geq n_0 \ \forall \Theta \hookrightarrow D\Theta_1 < D\tilde{\Theta}_2'$

$\Rightarrow \tilde{\Theta}_1$ эффективнее $\tilde{\Theta}_2'$

Проверим на регулярность

1) $p(x)$ - непр дифф на $\Theta > 0$

2) $$\frac{\partial}{\partial \Theta} \int_{-\infty}^{+\infty} p(x)\,dx = \frac{\partial}{\partial \Theta}\left(\frac{1}{\Theta}\int_0^{+\infty} e^{-\frac{x}{\Theta}} dx\right) =$$

$$= \frac{\partial}{\partial \Theta}\left(-\frac{1}{\Theta}\Theta e^{-\frac{x}{\Theta}}\right)\Big|_0^{+\infty} = \frac{\partial}{\partial \Theta}(1) = 0$$

$$\int_{-\infty}^{+\infty} \frac{\partial}{\partial \Theta} p(x)\,dx = \int_0^{+\infty} \frac{\partial}{\partial \Theta} \frac{e^{-x/\Theta}}{\Theta} dx = \int_0^{+\infty} \frac{\frac{x}{\Theta^2} e^{-\frac{x}{\Theta}} \cdot \Theta - e^{-\frac{x}{\Theta}}}{\Theta^2} dx =$$

$$= \int_0^{+\infty} \frac{x e^{-\frac{x}{\Theta}} - \Theta e^{-x/\Theta}}{\Theta^3} dx = \frac{1}{\Theta^2}\left(\int_0^{+\infty} \frac{x}{\Theta} e^{-\frac{x}{\Theta}} dx - \int_0^{+\infty} e^{-x/\Theta} dx\right) = 0$$

3) $$\ln p(x) = \ln \frac{e^{-x/\Theta}}{\Theta} = -\frac{x}{\Theta} - \ln \Theta$$

$$\left(\frac{2 \ln p(x)}{2\Theta}\right)^2 = \left(\frac{x}{\Theta^2} - \frac{1}{\Theta}\right)^2$$

$$I(\Theta) = M\left[\left(\frac{2 \ln p(x)}{2\Theta}\right)^2\right] = \int_0^{+\infty} \frac{(x-\Theta)^2}{\Theta^2} \frac{e^{-x/\Theta}}{\Theta^2} dx =$$

$= \frac{1}{\Theta^2}$ непр на $\Theta > 0$ и $I(\Theta) > 0$

$\Rightarrow$ модель регулярна

$\tilde{\theta}_1 = \bar{x}$ модель регулярна

$\tilde{\theta}_1$ - несмещ $D\tilde{\theta}_1 = \frac{\theta^2}{n}$ огр на $\forall$ компакте

$\Rightarrow$ оценка регулярна по дост усл

$\tilde{\theta}_2' = \frac{n(n-1)}{2n-1} x_{(2)}$ несмещ

$D\tilde{\theta}_2' = \theta^2 \left(\frac{2n^2-2n+1}{4n^2-4n+1}\right)$ - огр на $\forall$ компакте

$\Rightarrow \tilde{\theta}_2'$ - регулярна

Выполнены все условия нер-ва Крамера-Рао

$D\tilde{\theta}_1 \geq \frac{1}{3\frac{1}{\theta^2}} = \frac{\theta^2}{3}$

По Т. о единств. эфф оценки $\tilde{\theta}_1$ эффективнее